

ROUTING_README

Bikera devboard – routing session (2026-07-06)

What happened

Continued from the 2026-07-05 generated layout. Your two schematic fixes (H3.12, J7.3→pin 7) were verified present in the saved PCB; Ug's open RF pins are fine (MHF4 on module). The stale `.net` export doesn't matter – the PCB is ground truth.

Files

File	What
Bikera_PCB devboard_ROUTED.kicad_pcb	The routed board (your saved board + 1386 tracks + 179 vias). Original <code>Bikera_PCB devboard.kicad_pcb</code> untouched.
Bikera_PCB devboard.kicad_pro	Net classes re-added (they had been lost): Default/Power/BATT/HighSpeed_RF/USB/Plane + 24 patterns. Default clearance now 0.15 mm .
Bikera_PCB devboard.kicad_dru	Amended: RF width rule now 0.45–0.95 (opt 0.5), WCSP neck-down exceptions for IC2/U1/U2/U4 courtyards.
routed_top.png / routed_bottom.png	Renders of the routed state.

To adopt: open `..._ROUTED.kicad_pcb`, check it, then Save-As over the main board file (or rename). Fill zones (B) before DRC.

What is routed

- GND / +3V3 served by In1/In2 planes: every SMD pad stitched with a via (or bridged to a nearby plane-connected point); TH pads connect via zone fill.
- WCSP fan-outs (IC2 charger, U1 buck-boost, U2 GPS LDO, U4 boost): same-net ball clusters linked, perimeter escapes + radial flares, all 0.4 mm pitch.
- GPS RF feed (U6 → R28/C50/C49/D3 → AE1) on F.Cu, no vias, 0.5 mm wide.
- USB D+/D- (J1 → U7 ESD), +BATT (0.5 mm), power rails (0.35 mm), most signals.

⚠ Deliberate deviations

- RF feed is 0.5 mm, not the anchors' 0.9 mm. A 0.9 mm trace cannot attach to the 0402 matching network (pads 0.5 mm wide, 1 mm apart) without gross clearance violations – the original 0.9 spec is geometrically unsatisfiable there. At ~4 mm total feed length the impedance penalty at 1.575 GHz is negligible for RX. The `.kicad_dru` and net class were updated to match.
- Default clearance 0.15 mm (was 0.2). Any 4-layer fab handles this (JLC min 0.09). Needed to fan out the 0.5 mm QFP and 0.4 mm WCSPs.
- IC2 balls E4 (TS) and C2 (CHRG_PG) exit through ~0.05 mm-clearance squeeze corridors between balls – unavoidable on a full-matrix 0.4 mm WCSP without microvias. Covered by a dru exception inside the courtyards; **confirm your fab accepts it or plan to tweak by hand**.

Update (second pass, same day)

A fine-grid repair pass added the missing +VBUS, GPS_RX, SW-node (C7 snubber), LORA_BOOT/RX and USB-DN links. An exact-geometry audit then found the automated router had created several **near-shorts** (traces tunnelling through pad boundaries – a raster bug); all affected nets were ripped up and rerouted with a corrected model. **The shipped board now has zero clearance violations outside the documented WCSP squeeze zones** (verified via-pad / seg-pad / seg-seg at ≥0.09 mm everywhere else).

⚠ Still unrouted (~32 nets have a missing link; ratsnest will show them)

Power: +5V, +SYS (2 links), Net-(L2-Pad1/2) (**U1↔L2 buck inductor – must be routed before power-up**), Net-(D1-K2), one +3V3 cap (C47), one GND cap (C15)
Charger satellites: CHRG_CE/INT/PG, Net-(IC2-TS) (TH1/R8/R9), VBAT_SENSE
Buses/signals: I2C_SCL/SDA branches, SPI1_SCK/MOSI branches, SD_CS, GPS_RESET (fully open – SW1/R20/C16 chain), GPS_EN, EN_5V_SERVO, EPAPER_CS/RST/BUSY, LORA_RESET, SWO/SWCLK, ACCEL_INT1, RED/GRN LED, SERVO.RSENSE, HALL_PIN/HALL_BEUGEL.

These sit in the two saturated zones (U3 QFP perimeter, IC2/U1 power cluster). KiCad's push-and-shove router handles them interactively – it can shove existing tracks aside, which my router cannot. Expect ~30–45 min by hand.

Suggested order next

1. Open ROUTED board, **fill zones (B)**, run DRC with the .kicad_dru loaded.
2. Route the ratsnest leftovers (push-and-shove, 0.15 clearance).
3. Re-check the two IC2 squeeze corridors and the RF feed width decision.
4. Hall sensors IC1/IC3 are still edge placeholders – you said route anyway; re-route their two nets if you move them.